

Network Topologies — Guided Notes

Unit 1.2.1 | CompTIA Network+ N10-009 Obj. 1.2 | N10-008 1.2

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

In a _____ topology, all devices connect to a single shared cable called a _____. If that cable fails at any point, _____ devices lose connectivity.

Question 2

A ring topology passes a special _____ frame around the loop to control who can transmit. The main weakness of a single-ring design is that one _____ or _____ failure breaks the entire loop.

Question 3

In a star topology, all devices connect to a central _____. A single device failure affects only _____ device. The central device, however, is a _____ — if it fails, the entire segment goes down.

Question 4

In a full mesh topology with 10 devices, how many point-to-point links are required? Show the formula and the result.

Question 5

Explain the difference between a full mesh and a partial mesh. Why is partial mesh the more common real-world design?

Question 6

_____ topology combines two or more topology types. Give one real-world example: _____.

Question 7

Physical topology is the _____ layout of cables and devices. Logical topology is the path _____ actually takes. These two can differ — give an example of when they do.

Question 8 — Real World Application

REAL WORLD: An entire floor of a building loses network access simultaneously. No individual device issues are reported. Based on what you know about network topologies, what topology is most likely in use, what single component probably failed, and what design change would prevent this from happening again?

Download the answer key from the class server:

```
wget http://[SERVER]/resources/network-engineer/1-2-1-network-topologies/1-2-1-network-topologies-answer-key.pdf
```